

DECISION PROBLEMS IN PROPOSITIONAL/PREDICATE LOGIC

1. Is a propositional/predicate formula a *tautology/theorem*?

$$\stackrel{?}{\models} V \quad \text{or} \quad \stackrel{?}{\vdash} V$$

2. Is a propositional/ predicate formula a *logical/syntactic consequence* of a set of hypotheses?

$$U_1, \dots, U_n \stackrel{?}{\models} V \quad \text{or} \quad U_1, \dots, U_n \stackrel{?}{\vdash} V$$

In order to solve these two decision problems, *theorem proving methods* are applied.

Classification:

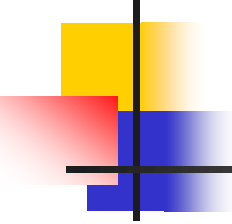
semantic *versus* syntactic proof methods

<i>semantic proof methods</i>	<i>syntactic proof methods</i>
propositional logic - the truth table method - CNF- conjunctive normal form propositional/predicate logic - the semantic tableaux method	propositional/predicate logic - the definition of deduction - the theorem of deduction and its reverse - the resolution method - the sequent calculus method

Classification:

direct *versus* refutation proof methods

<u>direct</u> <i>proof methods</i>	<u>refutation</u> <i>proof methods</i>
-they use directly the formula to be proved	- they model “ reductio ad absurdum ” principle (proof by contradiction) using the negation of the formula to be proved
propositional logic - the truth table method - CNF- conjunctive normal form propositional/predicate logic - the sequent calculus method - the definition of deduction - the theorem of deduction and its reverse	propositional/predicate logic - the semantic tableaux method; - the resolution method.



Semantic Tableaux Method

- It was proposed as a proof method for classical logics by R. Smullyan in 1968.
- Dedicated theorem provers: 3TAP, pTAP, leanTAP, Cassandra.
- It was easily adapted to *nonstandard logics* (modal, temporal, many-valued, non-monotonic).
- It is based on semantic considerations => ***semantic method.***
- Its basic aim is to decide ***consistency*** and to find all the models of a formula by decomposing the formula in sub-formulas.
- The ***validity*** of a formula **is proved by contradiction** =>
=> ***refutation method.***

Decomposition rules for *propositional formulas*

- **conjunctive formulas**, which are consistent only if both of its component sub-formulas are satisfied, are decomposed using α - **rules**

$$\begin{array}{c} A \wedge B \\ | \\ A \\ | \\ B \end{array}$$

$$\begin{array}{c} \neg (A \vee B) \equiv \neg A \wedge \neg B \\ | \\ \neg A \\ | \\ \neg B \end{array}$$

$$\begin{array}{c} \neg (A \rightarrow B) \equiv A \wedge \neg B \\ | \\ A \\ | \\ \neg B \end{array}$$

- **disjunctive formulas**, which are satisfied if one of its component sub-formulas is satisfiable(consistent) are decomposed using β - **rules**

$$\begin{array}{cc} A \vee B & \\ / \quad \backslash & \\ A & B \end{array}$$

$$\begin{array}{cc} \neg (A \wedge B) \equiv \neg A \vee \neg B & \\ / \quad \backslash & \\ \neg A & \neg B \end{array}$$

$$\begin{array}{cc} A \rightarrow B \equiv \neg A \vee B & \\ / \quad \backslash & \\ \neg A & B \end{array}$$

Recommendation:

- apply α - rules which keep one branch before β - rules which make a branching.



Decomposition rules for *predicate formulas*

γ - rules

$(\forall x)A(x)$	$\neg(\exists x)A(x)$
$A(c_1)$	$\neg A(c_1)$
$A(c_2)$	$\neg A(c_2)$
...	...
$A(c_n)$	$\neg A(c_n)$
$(\forall x)A(x)$	$\neg(\exists x)A(x)$

- used to decompose **universally quantified** formulas
- $c_1, c_2, \dots, c_n$ are all the parameters (constants) on that branch and they are used for the instantiation of A .
- if there is no constant on that branch, a new constant is introduced and used for instantiation.

δ - rules

$(\exists x)A(x)$	$\neg(\forall x)A(x)$
$A(c)$	$\neg A(c)$

- used to decompose **existentially quantified** formulas
- c is a new constant on the branch.

Recommendation:

- apply δ -rules (introduction of new constants) before γ -rules which use all the constants on that branch.

Construction of a semantic tableau

To a propositional/predicate formula U we can associate a semantic tableau, which is a binary tree having formulas in its nodes and it is built as follows:

1. the root of the tree is labeled with the initial formula;
2. every branch of the tree which contains a formula will be extended with a subtree according to the decomposition rule specific to its class;
3. the extension of a branch stops in the following cases:
 - a) if that branch contains a formula and its negation, the branch is marked as closed using the symbol $\otimes$;
 - b) if all the formulas on that branch are already decomposed or if by decomposing the formulas which are not decomposed yet, no new formulas are obtained.

Definitions:

1. A **branch** of the semantic tableau is called **closed** (marked by $\otimes$) if it contains a formula and its negation.
2. A **branch** is called **complete** if it is closed or all the formulas on that branch are already decomposed.
3. A **branch** is called **open** (marked by the symbol $\odot$) if it is complete but not closed.
4. A **semantic tableau** is called **closed** if all its branches are closed.
5. If a semantic tableau has at least one open branch, the **tableau** is called **open**.
6. A **semantic tableau** is called **complete** if all its branches are complete.

Remarks:

- A **closed branch** symbolizes **inconsistency** among the formulas on that branch.
- The set of formulas belonging to an open branch is consistent, so it has a model.
- A **formula is inconsistent** if and only if its **semantic tableau is closed**.
- A **formula is consistent** if and only if its **semantic tableau is open**, its open branches provide the models.

Models of a formula

➤ *propositional formula* U :

- by assigning the truth value T to all the literals belonging to a complete and open branch of the semantic tableau of U , a partial/complete model of U is obtained.
- if a propositional variable is not on that branch, we can assign to it either T or F , obtaining complete models.

➤ *predicate formula* U :

- an open branch of the semantic tableau associated to *formula* U provides a partial/complete model of U with the minimum number of elements in the domain of interpretation. Such a model $I = \langle D, m \rangle$ is a 'generic' one and it is built as follows:
 - the domain of interpretation D contains all the constants on that branch;
 - the value T is assigned to all the literals on that branch:
 - if $c_1, \dots, c_n \in D$, $P \in P_n$ (P is an n -ary predicate symbol) and
 - a) $P(c_1, \dots, c_n)$ belongs to the branch then $m(P)(c_1, \dots, c_n) = T$
 - b) $\neg P(c_1, \dots, c_n)$ belongs to the branch then $m(P)(c_1, \dots, c_n) = F$
- based on this generic model, more concrete models can be obtained.

Theorems

Semantic tableaux method – a *refutation* proof method

Theorem 1: Soundness and completeness of the semantic tableaux method

A propositional/predicate formula U is a tautology *if and only if* $\neg U$ has a closed semantic tableau.

Theorem 2:

Let $U_1, U_2, \dots, U_n, V$ be propositional/predicate formulas.

$U_1, U_2, \dots, U_n \models V$ *if and only if* there is a closed semantic tableau associated to the formula $U_1 \wedge U_2 \wedge \dots \wedge U_n \wedge \neg V$.

Propositional logic is decidable:

- the *semantic tableau* associated to a propositional formula *is always finite* therefore, we can always decide whether the formula is a tautology or not.

Predicate logic - undecidable

Theorem (Church 1936):

The problem of validity of a first-order formula is *undecidable*, but is *semi-decidable*. If a procedure *Proc* is used to check the validity of a first-order formula *U* we have the following cases:

- if *U* is a valid formula, **then** *Proc* ends with the corresponding answer.
- if the formula *U* is not valid, **then** *Proc* ends with the corresponding answer **or** *Proc* may never stop.

Predicate logic is not decidable, it is only semi-decidable:

- if the *semantic tableau* associated to the predicate formula $\neg U$ is *finite* then we can decide whether the formula *U* is a *tautology* (closed tableau for $\neg U$) or not (complete and open tableau for $\neg U$);
- if the *semantic tableau* of the predicate formula $\neg U$ is *infinite* then we cannot decide upon the validity of *U*.

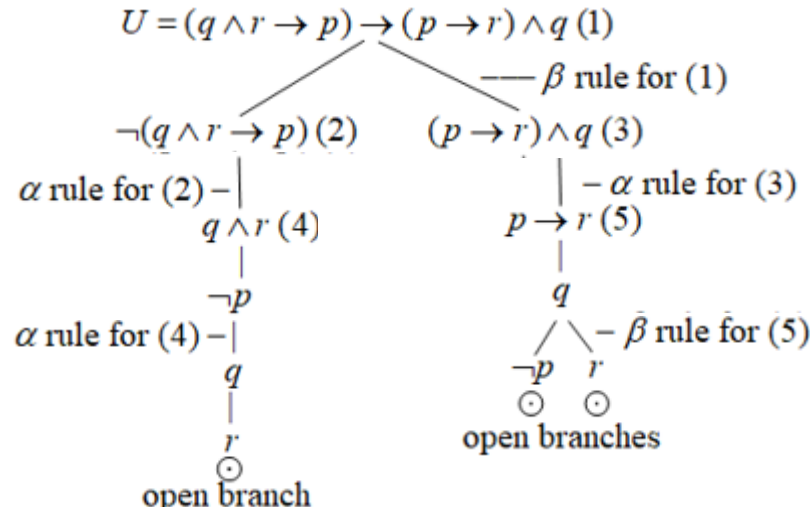
Theoretical results

Let U be a propositional formula or a closed predicate formula (all variables are bound):

- If U has a closed semantic tableau (all its branches are closed) then U is inconsistent.
- If U is a propositional/predicate formula and has an open semantic tableau (at least one open branch) then U is a consistent formula.
- If U is a closed predicate formula and has at least one infinite branch, the finite branches being closed then we cannot decide whether the formula is consistent or inconsistent.
- If $\neg U$ has a closed semantic tableau (all its branches are closed) then $\neg U$ is inconsistent and U is valid (tautology).
- The models of U are provided by the open branches of the semantic tableau of U .
- The anti-models of U are provided by the open branches of the semantic tableau of $\neg U$.

Example 1

Build a semantic tableau for $U = (q \wedge r \rightarrow p) \rightarrow (p \rightarrow r) \wedge q$ and decide the type of U . In case of consistency find all its models.



The semantic tableau is complete and open, having three open branches and thus U is consistent. $DNF(U) = (\neg p \wedge q \wedge r) \vee (q \wedge \neg p) \vee (q \wedge r) \equiv (q \wedge \neg p) \vee (q \wedge r)$

The simplified form of DNF was obtained by applying the absorption law. The left-most branch, corresponding to the first cube, is covered by the other two branches.

The branch represented by the cube $q \wedge \neg p$ provides the models: $i_1, i_2 : \{p, q, r\} \rightarrow \{T, F\}$ $i_1(p) = F, i_1(q) = T, i_1(r) = T$ $i_2(p) = F, i_2(q) = T, i_2(r) = F$	The branch represented by the cube $q \wedge r$ provides the models: $i_3, i_4 : \{p, q, r\} \rightarrow \{T, F\}$ $i_3(p) = T, i_3(q) = T, i_3(r) = T$ $i_4(p) = F, i_4(q) = T, i_4(r) = T$
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Because $i_1 = i_4$, U has three models: i_1 , i_2 and i_3 , $i_1(U) = i_2(U) = i_3(U) = T$

Having at least one model, U is a consistent formula, but it is not a tautology (does not have 8 models), therefore U is a contingent formula.

α - rule

$A \wedge B$

|
A

|
B

α - rule

$\neg(A \rightarrow B)$

|
A

|
 $\neg B$

β - rule

$A \rightarrow B$

/ \
 $\neg A$ B

Prove the validity of $U = (\neg q \rightarrow \neg p) \rightarrow ((\neg q \rightarrow p) \rightarrow q)$

We build a semantic tableau of $\neg U$.

$$\neg U = \neg((\neg q \rightarrow \neg p) \rightarrow ((\neg q \rightarrow p) \rightarrow q)) \quad (1)$$

- α rule for (1)

$$\neg q \rightarrow \neg p \quad (2)$$

$$\neg((\neg q \rightarrow p) \rightarrow q)(3)$$

– α rule for (3)

$$\neg q \rightarrow p \quad (4)$$

79

- β rule for (2)

9

ρ

closed branch $\otimes$

9

p

closed branch $\otimes$

⊗ closed branch

All three branches of the semantic tableau are closed, containing pairs of opposite literals: $(\neg q, q)$, $(\neg q, q)$, $(\neg p, p)$.

Therefore the formula $\neg U$ has no models so, it is an inconsistent formula.

We conclude that U is a tautology.

α - rule

$$\neg(A \rightarrow B)$$

$$A$$

$$\neg B$$

β - rule

$$A \rightarrow B$$

$\neg A$

$$B$$

Example 2



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Using the semantic tableaux method check whether the conclusion C is a logical consequence of the set of hypotheses: $\{H_1, H_2, H_3, H_4\}$.

Consider the following *hypotheses*:

H_1 . Mary will go to London this summer if both her friends Kate and Susan go.

H_2 . If Kate passes the English exam in May then she will go to London.

H_3 . Kate was in hospital from April until July and she didn't take the English exam.

H_4 . This summer Susan will go to London on a business trip.

and the *conclusion*:

C . Will Mary go to London this summer?

Notations for the propositional variables:

M – Mary will go to London

S – Susan will go to London

K – Kate will go to London

KE – Kate passed the English exam

$H_1 : K \wedge S \rightarrow M$

$H_2 : KE \rightarrow K$

$H_3 : \neg KE$

$H_4 : S$

$C : M$

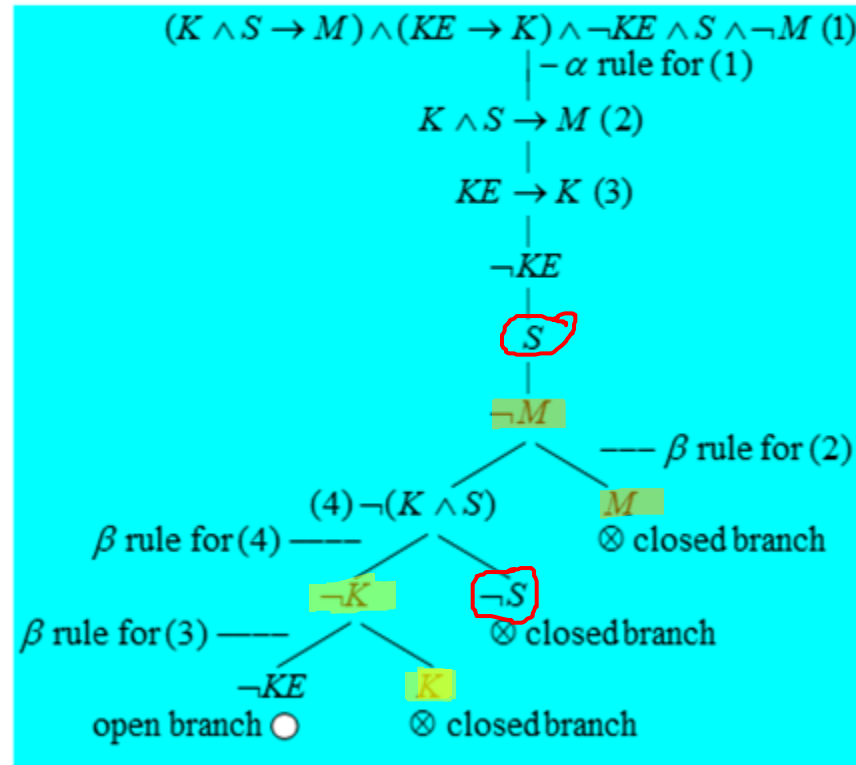
Example 2

The semantic tableau method is a refutation proof method, thus we have to negate the conclusion and use the **Theorem of soundness and completeness**:

$H_1, H_2, H_3, H_4 \models C$ if and only if

$H_1 \wedge H_2 \wedge H_3 \wedge H_4 \wedge \neg C$ has a closed semantic tableau.

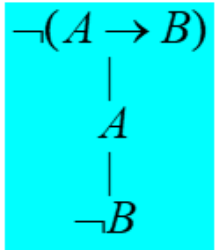
The semantic tableau corresponding to the conjunction of the hypotheses and the negation of the conclusion is depicted below.



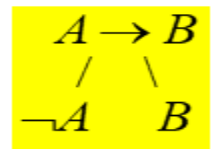
We have obtained a complete and open tableau with one open branch (the leftmost one) and three closed branches containing the following pairs of opposite literals: $(\neg K, K)$, $(S, \neg S)$, $(M, \neg M)$.

Therefore $H_1, H_2, H_3, H_4 \not\models C$ and based on the hypotheses we can't conclude that 'Mary will go to London this summer'.

α - rule



β - rule



Example 4. Build two different semantic tableaux for the formula:

$$U = (p \vee q) \wedge \neg(q \rightarrow r) \wedge (p \rightarrow q \wedge r).$$

Tableau 1

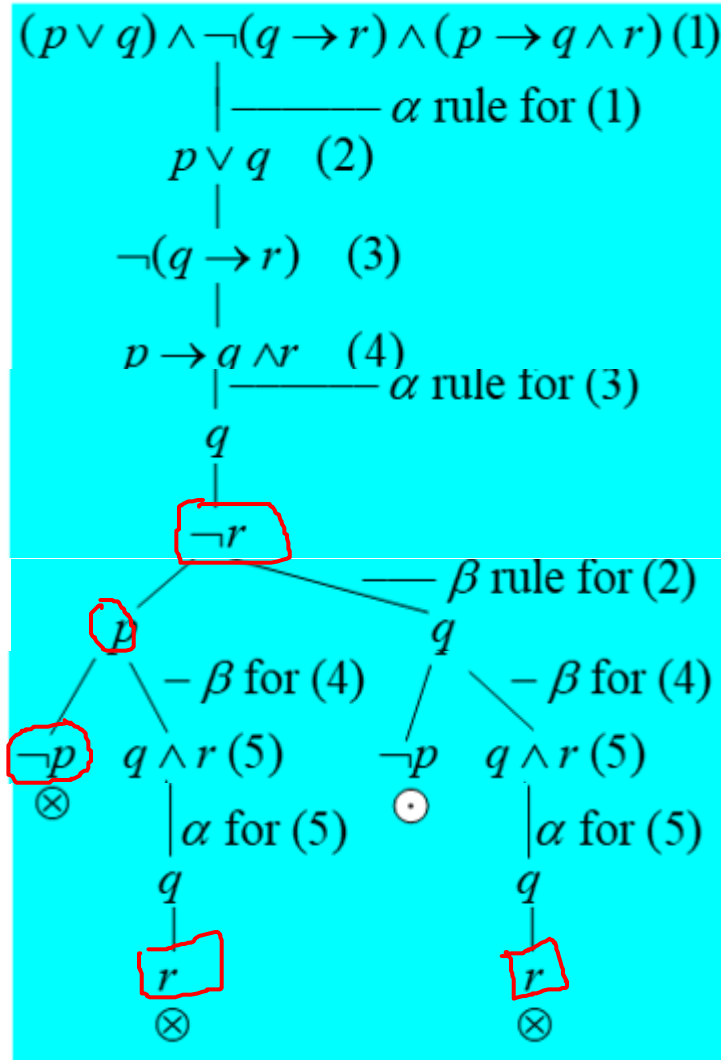
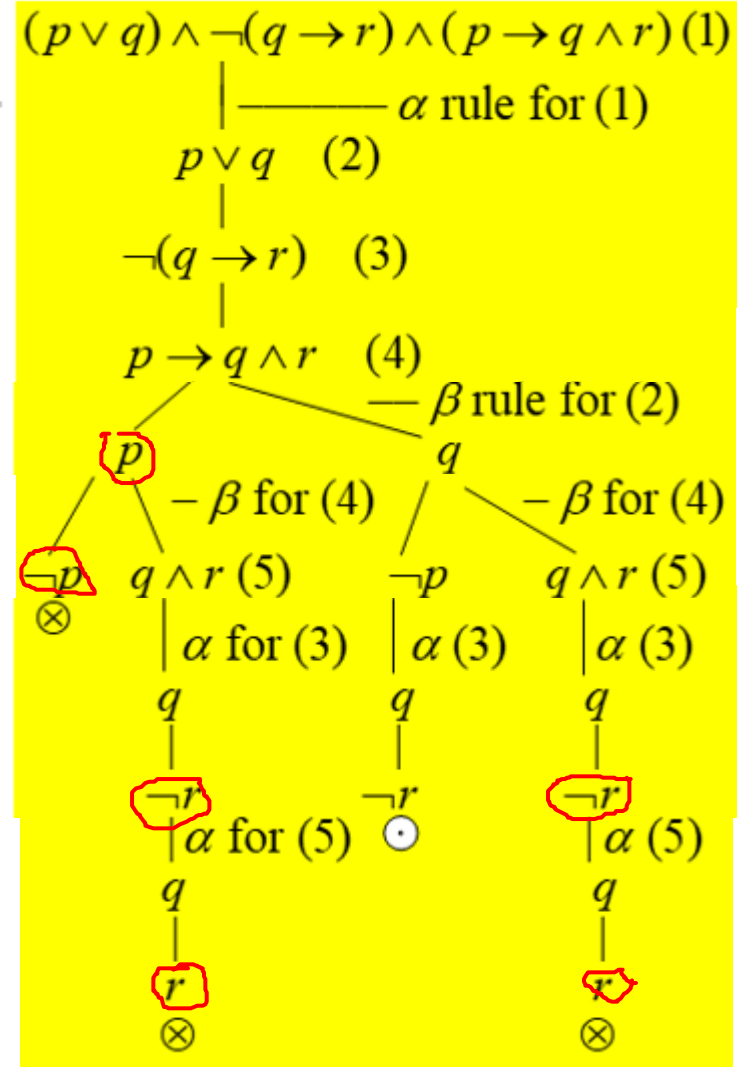


Tableau 2



α - rule

$$\frac{A \wedge B}{A}$$

$$\frac{A \wedge B}{B}$$

α - rule

$$\frac{\neg(A \rightarrow B)}{A}$$

$$\frac{\neg(A \rightarrow B)}{\neg B}$$

β - rule

$$\frac{A \rightarrow B}{\neg A \quad B}$$

β - rule

$$\frac{A \vee B}{A \quad B}$$

Famous Aristotle Quotes

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1. Knowing yourself is the beginning of all wisdom.
 2. It is the mark of an educated mind to be able to entertain a thought without accepting it.
 3. Patience is bitter, but its fruit is sweet.
 4. Without friends, no one would want to live, even if he had all other goods.
 5. It is during our darkest moments that we must focus to see the light.
 6. Virtue lies in our power, and similarly so does vice; because where it is in our power to act, it is also in our power not to act.
 7. Excellence is never an accident. It is always the result of high intention, sincere effort, and intelligent execution; it represents the wise choice of many alternatives - choice, not chance, and determines your destiny.

Example 5:

Check the **distributivity property of the *universal quantifier* over *implication***:

$$(\forall x)(P(x) \rightarrow Q(x)) \equiv (\forall x)P(x) \rightarrow (\forall x)Q(x)$$

We have that:

$$(\forall x)(P(x) \rightarrow Q(x)) \equiv (\forall x)P(x) \rightarrow (\forall x)Q(x) \text{ if and only if}$$

$$\models (\forall x)(P(x) \rightarrow Q(x)) \leftrightarrow ((\forall x)P(x) \rightarrow (\forall x)Q(x)) \text{ if and only if}$$

$$\models U_1 \text{ and } \models U_2 \text{ where:}$$

$$U_1 = (\forall x)(P(x) \rightarrow Q(x)) \rightarrow ((\forall x)P(x) \rightarrow (\forall x)Q(x)) \text{ and}$$

$$U_2 = ((\forall x)P(x) \rightarrow (\forall x)Q(x)) \rightarrow (\forall x)(P(x) \rightarrow Q(x)).$$

We shall prove that U_1 is a tautology and U_2 is not valid building the semantic tableaux of $\neg U_1$ and $\neg U_2$.

The following **theoretic result** is used:

$$\models U \text{ if and only if } \neg U \text{ has a closed semantic tableau.}$$

$$\neg U_1 = \neg((\forall x)(P(x) \rightarrow Q(x)) \rightarrow ((\forall x)P(x) \rightarrow (\forall x)Q(x))) \quad (1)$$

α - rule β - rule

$\neg(A \rightarrow B)$
 $\vdots$
 A
 $\vdots$
 $\neg B$

$A \rightarrow B$
 $\swarrow \quad \searrow$
 $\neg A \quad B$

γ - rules

$(\forall x)A(x)$ $\vdots$ $A(c_1)$ $\vdots$ $A(c_2)$ $\vdots$ $\dots$ $\vdots$ $A(c_n)$ $\vdots$ $(\forall x)A(x)$	$\neg(\exists x)A(x)$ $\vdots$ $\neg A(c_1)$ $\vdots$ $\neg A(c_2)$ $\vdots$ $\dots$ $\vdots$ $\neg A(c_n)$ $\vdots$ $\neg(\exists x)A(x)$
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δ - rules

$(\exists x)A(x)$ $\vdots$ $A(c)$	$\neg(\forall x)A(x)$ $\vdots$ $\neg A(c)$
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$(\forall x)(P(x) \rightarrow Q(x)) \quad (2)$

$\neg((\forall x)P(x) \rightarrow (\forall x)Q(x)) \quad (3)$

$(\forall x)P(x) \quad (4)$

$\neg(\forall x)Q(x) \quad (5)$

$\neg Q(a)$

$P(a)$

$(\forall x)P(x)$ copy of formula (4)

$P(a) \rightarrow Q(a) \quad (6)$

$(\forall x)(P(x) \rightarrow Q(x))$ copy of formula (2)

$\neg P(a)$
 closed branch $\otimes$

$Q(a)$
 $\otimes$ closed branch

The semantic tableau of $\neg U_1$ is closed, with two closed branches containing pairs of opposite literals: $(P(a), \neg P(a))$ and $(Q(a), \neg Q(a))$, therefore $\neg U_1$ is inconsistent and U_1 is valid.

γ - rules

$(\forall x)A(x)$	$\neg(\exists x)A(x)$
$\downarrow$	$\downarrow$
$A(c_1)$	$\neg A(c_1)$
$\downarrow$	$\downarrow$
$A(c_2)$	$\neg A(c_2)$
$\vdots$	$\vdots$
$\downarrow$	$\downarrow$
$A(c_n)$	$\neg A(c_n)$
$\downarrow$	$\downarrow$
$(\forall x)A(x)$	$\neg(\exists x)A(x)$

δ - rules

$(\exists x)A(x)$	$\neg(\forall x)A(x)$
$\downarrow$	$\downarrow$
$A(c)$	$\neg A(c)$

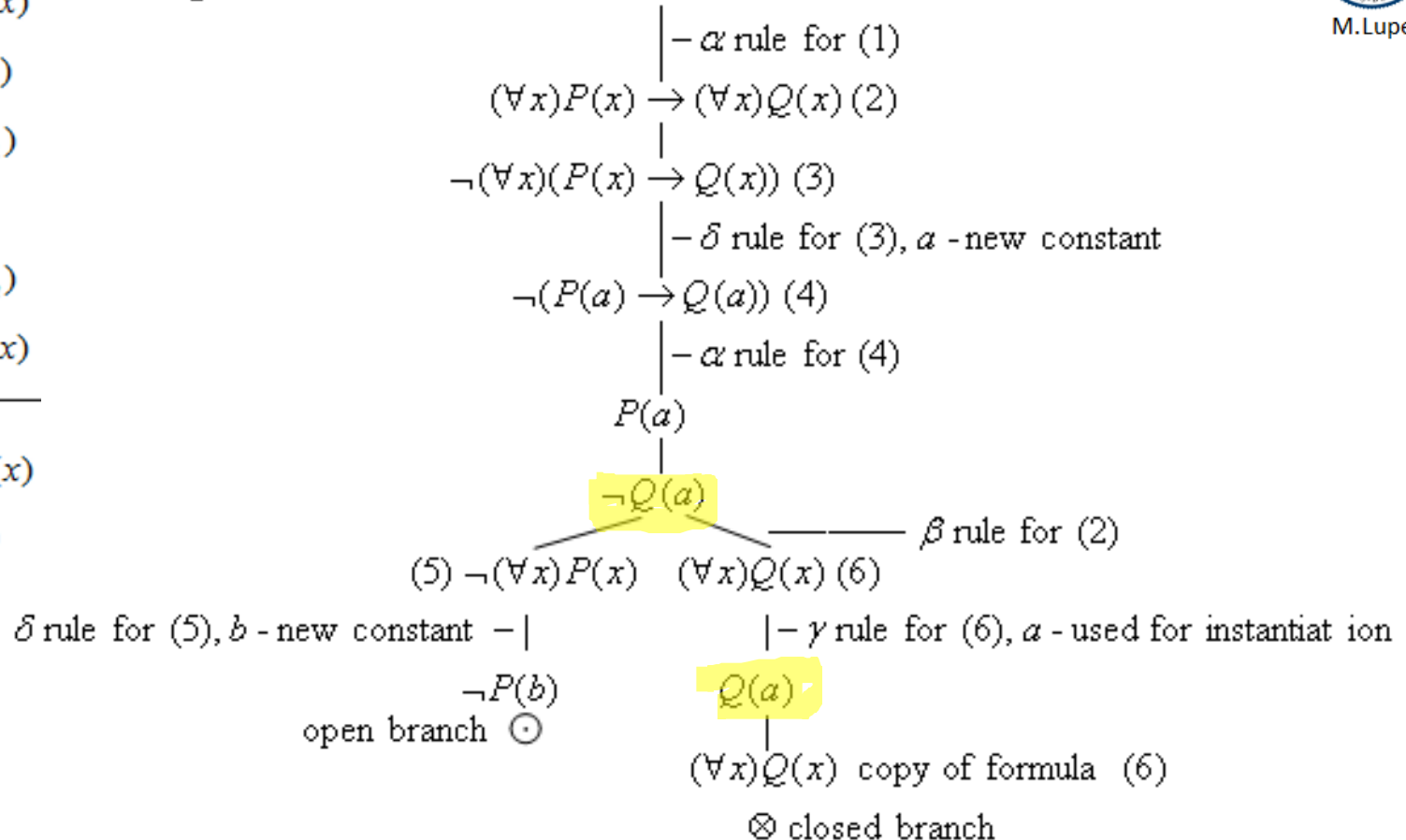
α - rule

$\neg(A \rightarrow B)$
$\downarrow$
A
$\downarrow$
$\neg B$

β - rule

$A \rightarrow B$
$\swarrow \quad \searrow$
$\neg A \quad B$

$$\neg U_2 = \neg(((\forall x)P(x) \rightarrow (\forall x)Q(x)) \rightarrow (\forall x)(P(x) \rightarrow Q(x))) \quad (1)$$



The semantic tableau of $\neg U_2$ is *finite, complete and open*, with one open branch and one closed branch, so $\neg U_2$ is *consistent* and U_2 is *not a tautology*.

Models of a predicate formula

➤ *predicate formula* U :

- an open branch of the semantic tableau associated to *formula* U provides a partial/complete model of U with the minimum number of elements in the domain of interpretation. Such a model $I = \langle D, m \rangle$ is a 'generic' one and it is built as follows:
 - the domain of interpretation D contains all the constants on that branch;
 - the value T is assigned to all the literals on that branch:
 - if $c_1, \dots, c_n \in D$, $P \in P_n$ (P is an n -ary predicate symbol) and
$$\begin{cases} \text{a) } P(c_1, \dots, c_n) \text{ belongs to the branch then } m(P)(c_1, \dots, c_n) = T \\ \text{b) } \neg P(c_1, \dots, c_n) \text{ belongs to the branch then } m(P)(c_1, \dots, c_n) = F \end{cases}$$
- based on this generic model, more concrete models can be obtained.

The models of formula $\neg U$ are the anti-models of U !!!!

Conclusion: We proved that $\models U_1$ and $\not\models U_2$ so,

*the universal quantifier is not distributive over implication,
it is only semi-distributive.*

The open branch contains the literals: $P(a)$, $\neg Q(a)$, $\neg P(b)$.

Find anti-models of the non-valid formula U_2 .

The open branch provides a partial model that covers two complete models of $\neg U_2$: I_1 and I_2 , which are anti-models of U_2 .

$I_1 = \langle D, m_1 \rangle$, where $D = \{a, b\}$ – the domain of interpretation

$m_1(P) : D \rightarrow \{T, F\}$, $m_1(P)(a) = T$, $m_1(P)(b) = F$,

$m_1(Q) : D \rightarrow \{T, F\}$, $m_1(Q)(a) = F$, $m_1(Q)(b) = T$

$v^{I_1}(\neg U_2) = T$ and $v^{I_1}(U_2) = F$.

$I_2 = \langle D, m_2 \rangle$, where $D = \{a, b\}$ – the domain of interpretation

$m_2(P) : D \rightarrow \{T, F\}$, $m_2(P)(a) = T$, $m_2(P)(b) = F$,

$m_2(Q) : D \rightarrow \{T, F\}$, $m_2(Q)(a) = F$, $m_2(Q)(b) = F$

$v^{I_2}(\neg U_2) = T$ and $v^{I_2}(U_2) = F$.

$I_1 = \langle D, m_1 \rangle$, where $D = \{a, b\}$ – the domain of interpretation

$m_1(P): D \rightarrow \{T, F\}$, $m_1(P)(a) = T$, $m_1(P)(b) = F$,

$m_1(Q): D \rightarrow \{T, F\}$, $m_1(Q)(a) = F$, $m_1(Q)(b) = T$

Concrete model based on the generic model I_1 :

$I_3 = \langle \Delta_3, m_3 \rangle$, $\Delta_3 = \{3, 4\}$

$m_3(P): \Delta_3 \rightarrow \{T, F\}$

$m_3(Q): \Delta_3 \rightarrow \{T, F\}$

$m_3(P)(x):$ "x is a prime"

$m_3(Q)(x):$ "x is a perfect square"

$m_3(P)(3) = T$, $m_3(P)(4) = F$

$m_3(Q)(3) = F$, $m_3(Q)(4) = T$

$\mathcal{V}^{I_3}(\neg \mathcal{U}_2) = T$, I_3 model of $\neg \mathcal{U}_2$

$\mathcal{V}^{I_3}(\mathcal{U}_2) = F$, I_3 anti-model of $\mathcal{U}_2$

Example 6: Prove the non-validity of $U = (\exists y)(\forall x)P(x,y)$

$\neg ((\exists y)(\forall x)P(x,y)) \equiv (\forall y)(\exists x) \neg P(x,y)$ (1)

| - γ rule for (1), c_0 - new constant

$(\exists x) \neg P(x, c_0)$ (2)

|

$(\forall y)(\exists x) \neg P(x,y)$ (3) copy of formula (1)

| - δ rule for (2), c_1 - new constant

$\neg P(c_1, c_0)$

| - γ rule for (3), c_1 is used for instantiation

$(\exists x) \neg P(x, c_1)$ (4)

|

$(\forall y)(\exists x) \neg P(x,y)$ (5) copy of formula (1)

| - δ rule for (4), c_2 - new constant

$\neg P(c_2, c_1)$

| - γ rule for (5), c_2 is used for instantiation

$(\exists x) \neg P(x, c_2)$ (6)

....

The semantic tableau of $\neg U$ is *infinite so*,
we cannot decide upon the validity of U .

A well known interpretation which falsifies U is
 $I = \langle D, m \rangle$, where the domain $D = \mathbb{N}$,
the set of all natural numbers and
 $m(P): \mathbb{N} \times \mathbb{N} \rightarrow \{T, F\}$, $m(P)(x, y): "x < y"$.

The formula $(\exists y)(\forall x)P(x,y)$ is evaluated under
the interpretation I as:

$$v^I(U) = (\exists y)_{y \in \mathbb{N}} (\forall x)_{x \in \mathbb{N}} "x < y"$$

and can be translated as:

"The biggest natural number exists",
which is obvious false, and thus U is not a tautology.



Famous Aristotle Quotes

8. Those who know, do. Those who understand, teach.
9. The worst form of inequality is to try to make unequal things equal.
10. We must no more ask whether the soul and body are one than ask whether the wax and the figure impressed on it are one.
11. You will never do anything in this world without courage. It is the greatest quality of the mind next to honor.
12. Educating the mind without educating the heart is no education at all.
13. We are what we repeatedly do. Excellence, then, is not an act, but a habit.
14. He who cannot be a good follower cannot be a good leader.
15. To avoid criticism say nothing, do nothing, be nothing.
16. Happiness belongs to the self-sufficient.